

Ch30: 3, 8, 13, 14, 24, 40, 43, 69

Basic score 4` + problem 17 2`+ problem 24 2`+ problem 43 2`

3. (a) Equation 30-8 leads to

$$\varepsilon = BLv = (2.4\text{T})(0.10\text{ m})(5.0\text{ m/s}) = 1.20\text{ V} . \quad 0.3\text{ point}$$

(b) By Lenz's law, the induced emf is clockwise. In the rod itself, we would say the emf is directed up the page. 0.2 point

(c) By Ohm's law, the induced current is $i = 1.20\text{ V}/0.40\ \Omega = 3.0\text{ A}$. 0.3 point

(d) The direction is clockwise. 0.2 point

(e) Equation 26-28 leads to $P = i^2R = 3.60\text{ W}$. 0.3 point

(f) From Eq. 29-2, we find that the force on the rod associated with the uniform magnetic field is directed rightward and has magnitude

$$F = iLB = (3.0\text{A})(0.10\text{ m})(2.4\text{T}) = 0.72\text{ N} .$$

To keep the rod moving at constant velocity, a leftward external force having that same magnitude must be continuously supplied to the rod.

0.2 point

(g) Using Eq. 7-48, we find the power associated with the force being exerted by the external agent:

$$P = Fv = (0.72\text{ N})(5.0\text{ m/s}) = 3.60\text{ W}, \quad 0.3\text{ point}$$

(h) which is the same as our result from part (e).

$$F = iLB = (3.0\text{A})(0.10\text{ m})(2.4\text{T}) = 0.72\text{ N} . \quad 0.2\text{ point}$$

8. Applying the loop theorem,

$$\varepsilon - L \left(\frac{di}{dt} \right) = iR ,$$

we solve for the (time-dependent) emf, with SI units understood:

$$\begin{aligned} \varepsilon &= L \frac{di}{dt} + iR = L \frac{d}{dt} (3.0 + 5.0t) + (3.0 + 5.0t)R = (6.0)(5.0) + (3.0 + 5.0t)(8.0) \\ &= (54 + 40t) . \end{aligned}$$

13. The flux over the loop cross section due to the current i in the wire is given by

$$\Phi = \int_a^{a+b} B_{\text{wire}} l dr = \int_a^{a+b} \frac{\mu_0 i l}{2\pi r} dr = \frac{\mu_0 i l}{2\pi} \ln\left(1 + \frac{b}{a}\right).$$

Thus,

$$M = \frac{N\Phi}{i} = \frac{N\mu_0 l}{2\pi} \ln\left(1 + \frac{b}{a}\right).$$

From the formula for M obtained above, we have

$$M = \frac{(150)(4\pi \times 10^{-7} \text{ H/m})(0.30 \text{ m})}{2\pi} \ln\left(1 + \frac{9.5}{1.0}\right) = 2.1 \times 10^{-5} \text{ H}.$$

14. The inductor prevents a fast build-up of the current through it, so immediately after the switch is closed, the current in the inductor is zero. It follows that

$$i_1 = \frac{\mathcal{E}}{R_1 + R_2} = \frac{100 \text{ V}}{10.0 \Omega + 20.0 \Omega} = 3.33 \text{ A}.$$

(b) $i_2 = i_1 = 3.33 \text{ A}$.

(c) After a suitably long time, the current reaches steady state. Then, the emf across the inductor is zero, and we may imagine it replaced by a wire. The current in R_3 is $i_1 - i_2$. Kirchhoff's loop rule gives

$$\begin{aligned}\mathcal{E} - i_1 R_1 - i_2 R_2 &= 0 \\ \mathcal{E} - i_1 R_1 - (i_1 - i_2) R_3 &= 0.\end{aligned}$$

We solve these simultaneously for i_1 and i_2 , and find

$$\begin{aligned}i_1 &= \frac{\mathcal{E}(R_2 + R_3)}{R_1 R_2 + R_1 R_3 + R_2 R_3} = \frac{(100 \text{ V})(20.0 \Omega + 30.0 \Omega)}{(10.0 \Omega)(20.0 \Omega) + (10.0 \Omega)(30.0 \Omega) + (20.0 \Omega)(30.0 \Omega)} \\ &= 4.55 \text{ A},\end{aligned}$$

(d) and

$$\begin{aligned}i_2 &= \frac{\mathcal{E} R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} = \frac{(100 \text{ V})(30.0 \Omega)}{(10.0 \Omega)(20.0 \Omega) + (10.0 \Omega)(30.0 \Omega) + (20.0 \Omega)(30.0 \Omega)} \\ &= 2.73 \text{ A}.\end{aligned}$$

(e) The left-hand branch is now broken. We take the current (immediately) as zero in that branch when the switch is opened (that is, $i_1 = 0$).

(f) The current in R_3 changes less rapidly because there is an inductor in its branch. In fact, immediately after the switch is opened it has the same value that it had before the switch was opened. That value is $4.55 \text{ A} - 2.73 \text{ A} = 1.82 \text{ A}$. The current in R_2 is the same but in the opposite direction as that in R_3 , that is, $i_2 = -1.82 \text{ A}$.

A long time later after the switch is reopened, there are no longer any sources of emf in the circuit, so all currents eventually drop to zero. Thus,

(g) $i_1 = 0$, and

(h) $i_2 = 0$.

24. (a) $t/\tau_L = Rt/L = (0.1\text{s})(12\Omega)/(2\text{H}) = 0.6$.

The rate at which the energy is being stored in the inductor is

$$\frac{dU_B}{dt} = \frac{d\left(\frac{1}{2}Li^2\right)}{dt} = Li \frac{di}{dt} = L \left(\frac{\mathcal{E}}{R} (1 - e^{-t/\tau_L}) \right) \left(\frac{\mathcal{E}}{R} \frac{1}{\tau_L} e^{-t/\tau_L} \right) = \frac{\mathcal{E}^2}{R} (1 - e^{-t/\tau_L}) e^{-t/\tau_L} = 2.06 \times 10^2 \text{ W}.$$

1 point 过程结果各 0.5 分

(b) the rate at which the resistor is generating thermal energy is

$$P_{\text{thermal}} = i^2 R = \frac{\mathcal{E}^2}{R^2} (1 - e^{-t/\tau_L})^2 R = \frac{\mathcal{E}^2}{R} (1 - e^{-t/\tau_L})^2.$$

At $t = 0.10 \text{ s}$, this yields $P_{\text{thermal}} = 1.70 \times 10^2 \text{ W}$.

0.5 point 过程 0.3 结果 0.2

(c) By energy conservation, the rate of energy being supplied to the circuit by the battery is

$$P_{\text{battery}} = P_{\text{thermal}} + \frac{dU_B}{dt} = 3.76 \times 10^2 \text{ W}.$$

0.5 point 过程 0.3 结果 0.2

We note that this result could alternatively have been found from

$$P_{\text{battery}} = \mathcal{E}i = \frac{\mathcal{E}^2}{R} (1 - e^{-t/\tau_L}) = 3.76 \times 10^2 \text{ W}.$$

40. (a) We assume the flux is entirely due to the field generated by the long straight wire (which is given by Eq. 29-17). We integrate according to Eq. 30-1, not worrying about the possibility of an overall minus sign since we are asked to find the absolute value of the flux.

$$|\Phi_B| = \int_{r-b/2}^{r+b/2} \left(\frac{\mu_0 i}{2\pi r} \right) (a dr) = \frac{\mu_0 i a}{2\pi} \ln \left(\frac{r+b/2}{r-b/2} \right).$$

When $r=1.5b$, we have

$$|\Phi_B| = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(6.9 \text{ A})(0.022 \text{ m})}{2\pi} \ln(2.0) = 2.1 \times 10^{-8} \text{ Wb}.$$

(b) Implementing Faraday's law involves taking a derivative of the flux in part (a), and recognizing that $dr/dt = v$. The magnitude of the induced emf divided by the loop resistance then gives the induced current:

$$\begin{aligned} i_{\text{loop}} &= \left| \frac{\mathcal{E}}{R} \right| = -\frac{\mu_0 i a}{2\pi R} \left| \frac{d}{dt} \ln \left(\frac{r+b/2}{r-b/2} \right) \right| = \frac{\mu_0 i a b v}{2\pi R [r^2 - (b/2)^2]} = \frac{\mu_0 i a b v}{2\pi R (2b^2)} \\ &= \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(6.9 \text{ A})(0.022 \text{ m})(0.0080 \text{ m})(3.2 \times 10^{-3} \text{ m/s})}{2\pi (4.0 \times 10^{-4} \Omega) [2(0.0080 \text{ m})^2]} \\ &= 1.52 \times 10^{-5} \text{ A}. \end{aligned}$$

43. (a) Consider a (thin) strip of area of height dy and width $\ell = 0.020 \text{ m}$. The strip is located at some $0 < y < \ell$. The element of flux through the strip is

$$d\Phi_B = B dA = (5.0t^2 y)(\ell dy) \quad 0.5 \text{ point}$$

where SI units (and 2 significant figures) are understood. To find the total flux through the square loop, we integrate:

$$\Phi_B = \int d\Phi_B = \int_0^\ell (5.0t^2 y \ell) dy = 5.0t^2 \frac{\ell^3}{2}. \quad 0.5 \text{ point}$$

Thus, Faraday's law yields

$$|\mathcal{E}| = \left| \frac{d\Phi_B}{dt} \right| = 5.0t\ell^3 = (5.0)(2.5)(0.03)^3 = 3.4 \times 10^{-4} \text{ V} \quad 0.5 \text{ point}$$

(b) Its "direction" (or "sense") is clockwise, by Lenz's law. 0.5 point

69. First we write $\Phi_B = BA \cos \theta$. We note that the angular position θ of the rotating coil is measured from some reference line or plane, and we are implicitly making such a choice by writing the magnetic flux as $BA \cos \theta$ (as opposed to, say, $BA \sin \theta$). Since the coil is rotating steadily, θ increases linearly with time. Thus, $\theta = \omega t$ if θ is understood to be in radians (here, $\omega = 2\pi f$ is the angular velocity of the coil in radians per second).

Since the area of the rectangular coil is $A = (0.500 \text{ m}) \times (0.300 \text{ m}) = 0.150 \text{ m}^2$, Faraday's law leads to

$$\varepsilon = -N \frac{d(BA \cos \theta)}{dt} = -NBA \frac{d \cos(2\pi ft)}{dt} = NBA 2\pi f \sin(2\pi ft)$$

which means it has a voltage amplitude of

$$\varepsilon_{\max} = 2\pi f NAB = 2\pi \left(\frac{1278}{60\text{s}} \right) (100 \text{ turns}) (0.15 \text{ m}^2) (4.3 \text{ T}) = 8.63 \times 10^3 \text{ V} .$$